

Business Summary Report: Predictive Insights for Collections Strategy

1. Summary of Predictive Insights

- Recent payment history (Month_1) was the strongest early indicator of delinquency. Customers with a recent late or missed payment showed a higher likelihood of future delinquency.
- Higher credit utilization was associated with a slightly greater risk of delinquency, suggesting that customers using a larger share of their available credit may require closer monitoring.
- Income showed a weak but noticeable relationship with delinquency, indicating that repayment capacity should be considered alongside payment behaviour and credit utilisation.

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Early payment behaviour predicts future risk	Customers with recent late or missed payments	Month_1, Credit_Utilization, Income	Supports early intervention and reduces delinquency

2. Recommendation Framework

Restated Insight: Recent late or missed payments are the strongest early indicator of customer delinquency.

Proposed Recommendation: Implement an automated early-warning and payment reminder programme for customers with a late or missed payment in Month_1.

Specific: Target customers who record a late or missed payment in Month_1.

Measurable: Reduce new delinquent accounts by at least 10% within six months.

Actionable: Send automated reminders and proactively contact high-risk customers.

Relevant: Focuses on the strongest risk factor identified during analysis.

Time-bound: Implement and evaluate the programme over the next six months.

Justification and Business Rationale: Early intervention after the first missed payment can encourage timely repayments before financial difficulties become more severe. This supports Geldium's objective of reducing delinquency while improving customer retention and credit risk management.

3. Ethical and Responsible AI Considerations

- Review model performance across demographic groups such as Employment_Status and Location to identify potential bias. If disparities are detected, review the training data, feature selection, and model before deployment.

- The model should support, not replace, human decision-making. Predictions should be explained as risk estimates based on customer financial and payment history rather than automatic lending decisions.
- Customer data should be handled securely, and the model should be monitored regularly to maintain fairness, transparency, accountability, and regulatory compliance.